

InternSpark Internship: Task 1 Report

Task Name: Customer Behaviour Analysis

Candidate Name: S SUDEEP

Domain: Data Analytics Intern

Project GitHub Link: <https://github.com/sudeep291/InternSpark-Internship/tree/main/Task-1-Customer-Behavior>

1. Executive Summary

The goal of this project was to analyze e-commerce customer transaction history to identify key behavioral segments, study product preferences, and flag immediate customer retention risks for Alfido Tech.

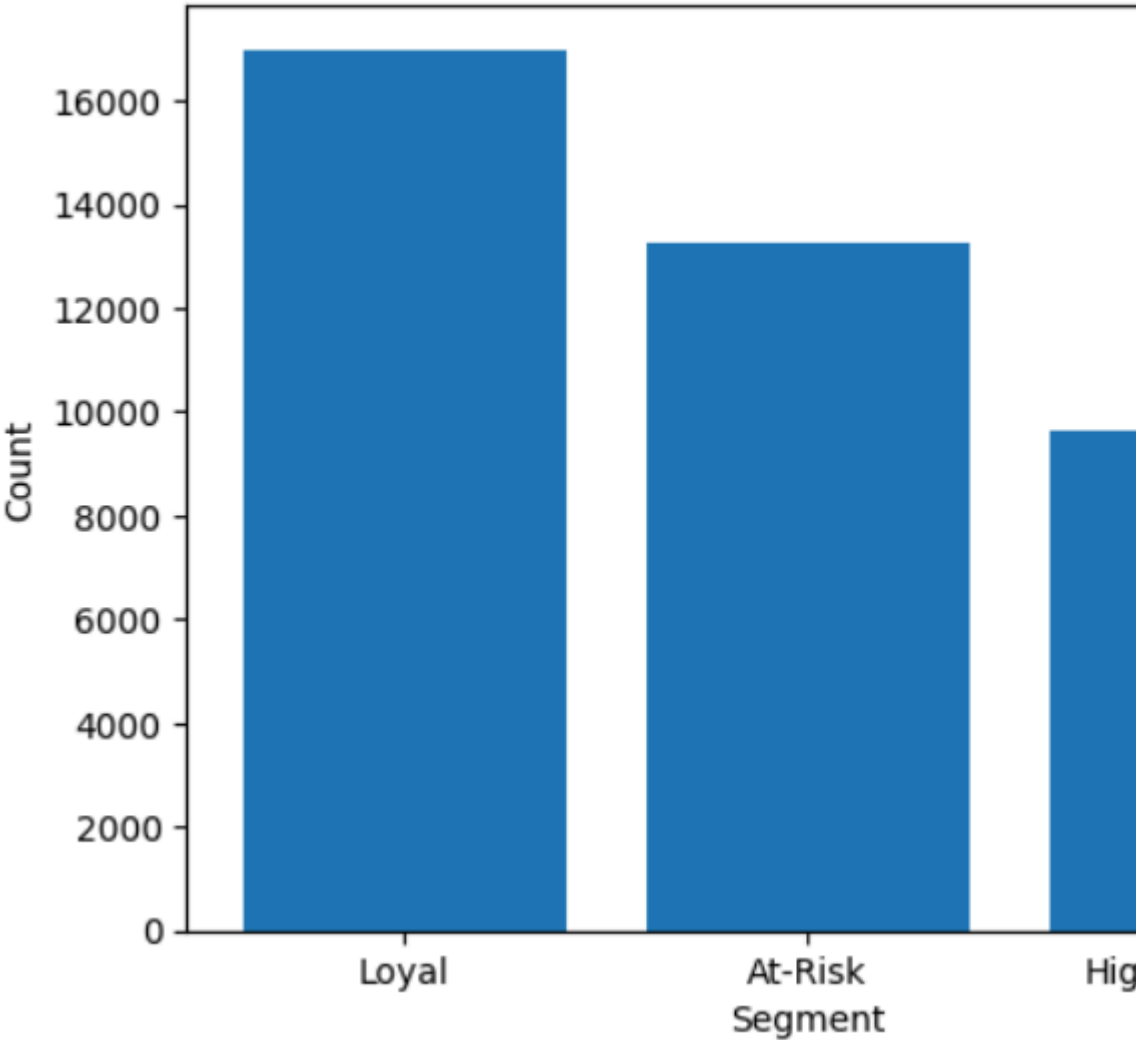
Through standard quantile engineering metrics, the active customer base was categorized into three distinct business profiles:

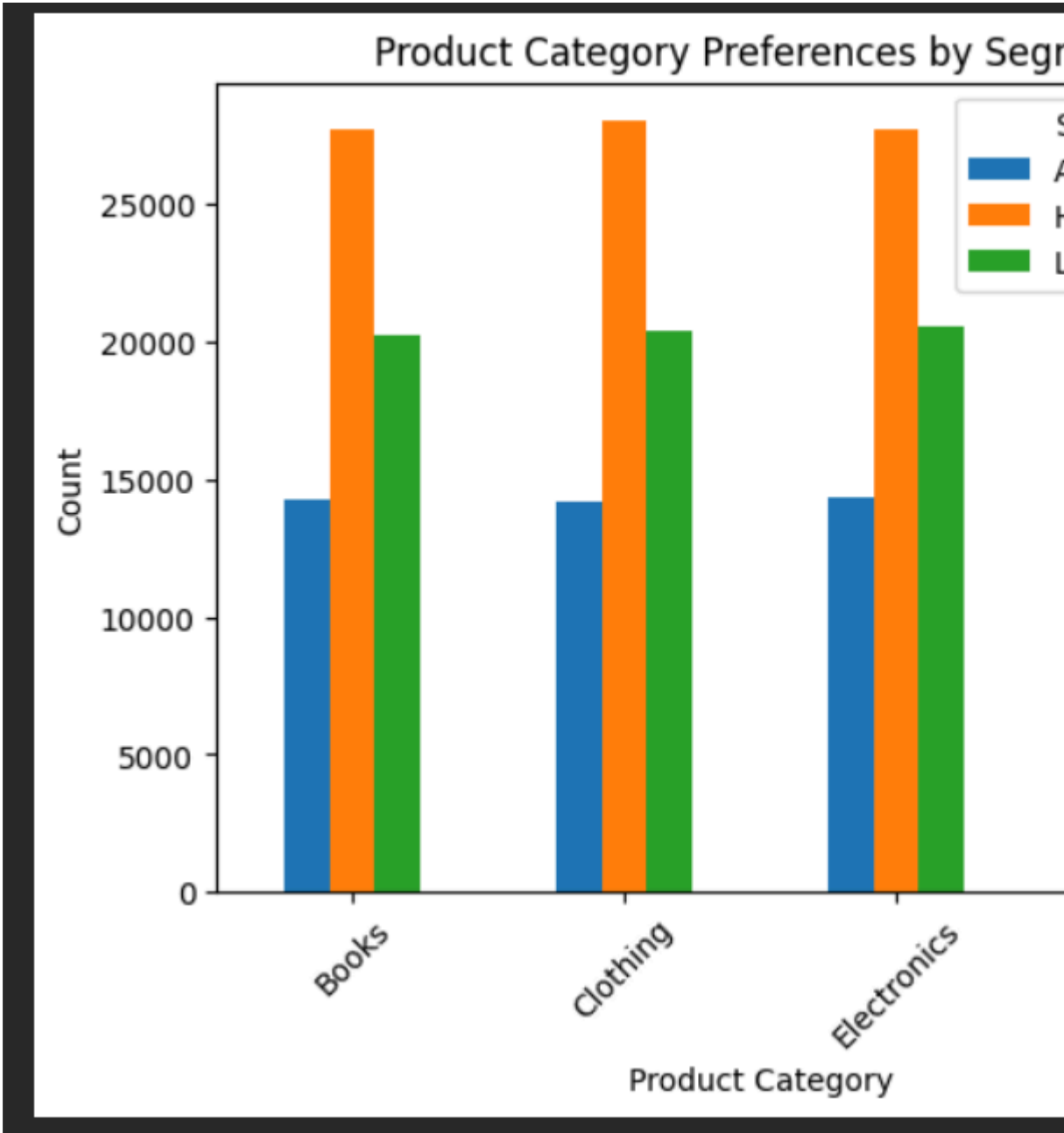
- **High-Spenders:** Top-tier revenue drivers demonstrating high transaction volume and premium category purchasing habits.
 - **Loyal Customers:** Highly active and consistent mid-to-high frequency buyers with predictable check-out habits.
 - **At-Risk Customers:** Lapsed accounts exhibiting large days-since-last-purchase intervals. They represent an immediate churn liability.
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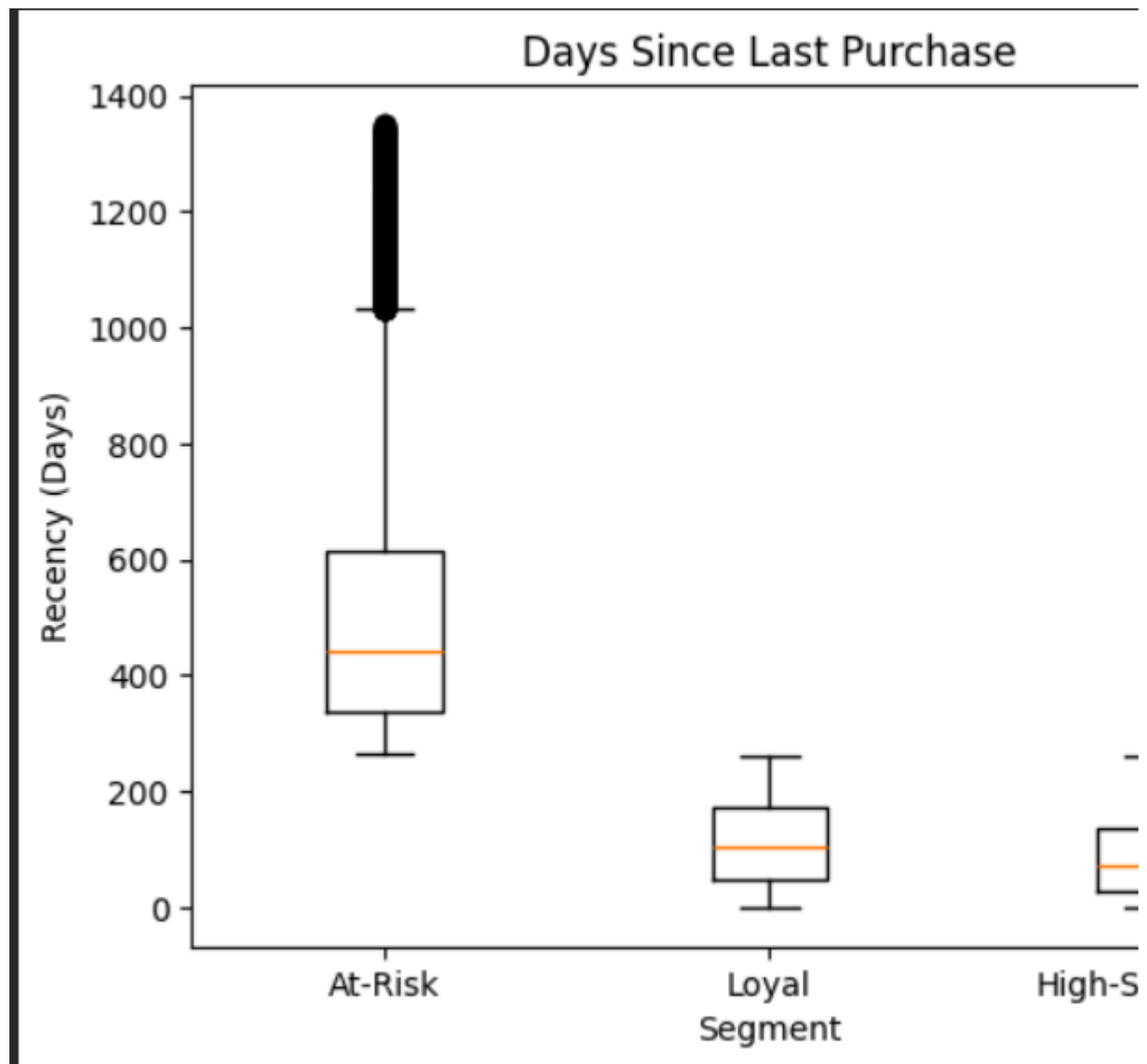
2. Visualizations & Verification Proof

Below is the verification proof displaying the three simple Matplotlib visualizations tracking customer breakdown, product categories, and retention trends.

Customer Segments Breakdown







3. 5 Actionable Recommendations for Alfido Tech

1. **Win-Back Promotions:** Automate targeted re-engagement email campaigns offering high-value discounts to the engineered **At-Risk** customer cluster to minimize platform churn.
2. **Tiered Loyalty Programs:** Establish a point-based milestone reward system for **Loyal** users to further maximize their weekly purchase frequency.
3. **VIP Tier Incentives:** Roll out premium customer perks, such as free express shipping, to maximize the long-term customer lifetime value (LTV) of **High-Spenders**.

4. **Proactive Recency Alerts:** Implement an internal metric monitor to flag customer accounts automatically when their days-since-last-purchase threshold crosses 60 days.
 5. **Smart Product Bundles:** Package frequently co-purchased items from overlapping categories (e.g., Electronics and Books) during the checkout process to increase average basket size.
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